NEXUS-AI: AI INSIGHTS & NOVELTY ENGINE

Technical Architecture Specification: Dual-Mode LLM + Heuristic Engine for Research Evaluation

Service Implementation Analysis: OllamaSummarizationService.java (Lines 111-153) & Patent Claims Pipeline

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1. DUAL-MODE AI ENGINE ARCHITECTURE

The Nexus-AI platform features a fault-tolerant, privacy-preserving Dual-Mode AI Insights Engine implemented in OllamaSummarizationService.java. The engine evaluates student research submissions, extracts technical problem statements, scores inventive novelty, and provides a preliminary patentability screening for institutional reviewers.

Mode 1: Primary Local LLM (Online)

- Engine: Ollama API running Llama 3.2 (2.0 GB)
- Base URL: http://localhost:11434/api/generate
- Parameters: Temperature=0.2, Top_P=0.9, Predict=800
- Persona: Principal Patent Examiner & Senior Fellow

Mode 2: Heuristic Fallback (Offline)

- Trigger: Ollama offline, timeout (>4s), or connection error
- Lexical Engine: 24 Weighted Technical Indicators
- Algorithm: Clamped Novelty Scoring (76% to 96%)
- Categorization: Utility Patent vs Software Copyright

2. LEXICAL ANALYZER & 24 TECHNICAL INDICATORS

Lines 114–122 of OllamaSummarizationService.java declare a curated dictionary of 24 domain keywords representing advanced engineering complexity. The engine scans the combined text of the submitted abstract and proposal document to compute keyword frequency hits:

algorithm	neural	framework	real-time	optimization	low-latency
autonomous	novel	sensor	architecture	distributed	blockchain
transformer	pipeline	synthesis	throughput	quantum	protocol
embedded	biometric	telemetry	model	inference	dataset

3. NOVELTY SCORING & IP CATEGORIZATION LOGIC

Lines 124–128 execute the patentability heuristics that determine whether an undergraduate project qualifies for formal Intellectual Property registration:

```
int noveltyScore = Math.min(96, Math.max(76, 75 + (keywordHits * 3)));
```

- Minimum Baseline Score: 76% (Ensures standard collegiate baseline for verified academic submissions)
- Scaling Factor: +3% per confirmed technical indicator hit from the 24-indicator dictionary
- Ceiling Cap: 96% (Reserves 100% perfection for post-examination patent office clearance)
- Classification Rule: noveltyScore >= 88% -> PATENT (Utility Invention). If software/app keywords match -> COPYRIGHT (Software & System Logic), else PATENT (Design & Utility).

4. THE 5 STRUCTURED RESEARCH EVALUATION VECTORS

Both the Ollama Llama 3.2 system prompt and the heuristic engine generate structured evaluations strictly divided into five standardized dimensions:

Vector 1: Core Problem & Novel Hypothesis

Identifies technical bottlenecks within the specified department (CSE, ECE, AI&DS, etc.) and domain. Formulates the project's hypothesis to reduce latency, automate workflows, and enhance operational throughput.

Vector 2: Methodology & Technical Innovations

Details architecture integration, multi-stage operational pipelines, localized edge/cloud optimizations, and empirical validation against standard benchmark datasets.

Vector 3: Patentability & Novelty Index

Outputs the quantitative Novelty Score (76% to 96%), assigns legal IP categorization (Utility Patent vs. Software Copyright), and articulates the non-obvious inventive step.

Vector 4: Prior-Art Vulnerabilities & Deficiencies

Examines generic market approaches and highlights potential claim-breadth challenges. Recommends specific hardware/software integration parameters to strengthen Patent Claim 1 against citations.

Vector 5: Commercialization & Translation Trajectory

Estimates the Technology Readiness Level (TRL 4: Component validation in laboratory) and projects industrial applicability for enterprise pilots, smart campuses, and startup spin-offs.

5. AI PATENT CLAIMS STUDIO & IPO FORM 2 SYNTHESIS

When a project reaches 'READY_FOR_IP' status, PatentClaimsService.java transforms the AI insights into formal Indian Patent Office (IPO) Form 2 documentation:

- **IPC Semantic Mapping:** Maps AI/ML to G06N 3/08; IoT/Telemetry to H04L 67/12; Healthcare to A61B 5/00; Robotics to B25J 9/16.
- **Independent Claim 1:** Drafts comprehensive apparatus claim: telemetry acquisition interface, distributed controller, optimized execution engine, deterministic parity verification layer.
- **Dependent Claims 2-5:** Elaborates dynamic adaptive windowing, cryptographic token parity hashing, self-healing sub-task rerouting (<250ms), and non-transitory storage medium.
- **Prior-Art Radar:** Evaluates synthetic prior-art collisions against USPTO, IPO India, and EPO patent citations with overlap percentages and technical differentiators.

6. ACADEMIC & TECHNICAL ATTESTATION

VERIFIED TECHNICAL FIDELITY:

The AI Insights Engine and Patent Claims Studio implementations represent a viable, state-of-the-art academic contribution for undergraduate research commercialization. The dual-mode architecture guarantees 100% uptime during academic evaluation periods even in air-gapped or resource-constrained lab environments.

AI Research Fellow / Candidate	**Faculty Supervisor (Mrs. Manju M)**	**Head of Department (Dr. Gomathi R)**
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